

WEEK – 16

DURATION: 27 April 2026 – 01 May 2026

DAY 1: 27 April 2026

1. TASKS WORKED ON

- **AI Chatbot UI & Backend Integration (“mb-ai-software” Project):** Integrated the chatbot frontend with the FastAPI backend and implemented core chat functionality, including message handling, typing effects, and loading indicators. Also enhanced the UI, resolved runtime issues, and improved response presentation through structured and user-friendly rendering.
- **Backend Learning – Node.js & Express Basics:** Learned the fundamentals of backend development, including servers, databases, and Node.js concepts. Also explored Node.js architecture, compared it with Java, and built a basic Express.js server to understand backend routing and application flow.

2. LEARNING AND KNOWLEDGE. GAINED

Today, I gained practical understanding of frontend-backend communication, chatbot integration, and Node.js backend fundamentals. I also improved my UI/UX skills by implementing interactive chatbot features and response handling.

3. PLAN FOR NEXT WORKING DAY

- Learn advanced concepts related to Node.js and Express.
- Improve the chatbot UI further by making it more interactive and visually enhanced.

DAY 2: 28 April 2026

1. TASKS WORKED ON

- ***Trailer Tracking System – Theme Toggle Implementation:*** Implemented theme toggle functionality using ThemeContext and improved the application's light and dark mode support. Also refined the sidebar and header styling to maintain a consistent, visually appealing, and user-friendly interface.
- ***Trailer Module Enhancement:*** Added new fields including Make, Model, Purchase Amount, and Selling Amount in the Trailer Info form. Updated form state handling and ensured proper data integration.
- ***Customer Module Update:*** Updated the Customer module by adding new fields such as Company Name, Address, and EIN Number to align frontend functionality with backend requirements. Improved form handling for both Individual and Company user types.

2. LEARNING AND KNOWLEDGE GAINED

Today, I learned how to implement theme switching functionality using React Context and understood how to maintain component independence from theme-specific styling.

I also improved my understanding of dynamic form handling, conditional field rendering, and maintaining visual consistency across different modules within a large application.

3. PLAN FOR NEXT WORKING DAY

- Continue working on the “MB-AI Software” project.
- Continue development work on the “NetSquare Internship Portal”.

DAY 4: 30 April 2026

1. TASKS WORKED ON

- **Chatbot Interface Development & Integration:** Developed and enhanced the chatbot onboarding page by improving its layout, styling, and overall user experience before users enter the main chat interface.
 - Integrated the frontend with backend APIs and implemented WebSocket-based real-time communication to enable smooth and responsive chatbot interactions.
 - Resolved chat data rendering issues by implementing a more flexible response handling system, allowing different types of backend data to be displayed correctly within the chat interface.
 - Improved application stability through better error handling, fallback states, and response validation, while also optimizing the overall data flow between the backend and frontend.

2. LEARNING AND KNOWLEDGE GAINED

Today, I learned how frontend and backend systems communicate using both REST APIs and WebSocket-based real-time communication. I also gained practical understanding of handling dynamic AI-generated responses and learned why flexible rendering systems are important in chatbot applications.

Additionally, I improved my debugging and optimization skills by analyzing data flow issues and implementing adaptable rendering strategies for multiple types of backend responses.

3. PLAN FOR NEXT WORKING DAY

- Continue enhancing the chatbot system with improved rendering strategies.
- Explore structured JSON-based response handling from the backend.
- Continue working on “MB-AI-Software” and “NetSquare Internship Portal”.

DAY 5: 01 May 2026

1. TASKS WORKED ON

- ***AI Chatbot – UI Enhancement & Feature Development:*** Enhanced the chatbot UI by improving its design, layout, animations, and overall user experience. Implemented features such as a profile section, chat history, and expand/collapse functionality, while also resolving theme-related issues and integrating the frontend with backend APIs for dynamic data handling.
- ***NetSquare Internship – Form Validation Enhancement:*** Implemented form validation features to ensure proper user input handling, improve data integrity, and enhance overall user experience.
- ***Backend Learning – Node.js Core Modules:*** Learned the fundamentals of the Node.js File System (FS) and Path modules, including file handling, directory management, and efficient path resolution in backend applications.

2. LEARNING AND KNOWLEGE GAINED

Today, I improved my understanding of UI enhancement, frontend–backend integration, and dynamic application features such as chat history and animations. I also gained practical knowledge of API communication, form validation, and backend file handling using Node.js.

3. PLAN FOR NEXT WORKING DAY

- Continue working on the “MB-AI Software” project.
- Learn additional backend development concepts and advanced Node.js topics.